MIL-P-46658A(MU)
21 September 1967
SUPERSEDING
MIL-P-46658(CmlC)
26 April 1962

MILITARY SPECIFICATION

PROJECTILE LOAD, 5 INCH 54 CALIBER,

WP, MK14; FILLING ASSEMBLY

1. SCOPE

1.1 This specification covers the white phosphorus (WP) filling and assembly of a projectile load without explosive components, for use in the 5 inch 54 caliber smoke projectile.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-F-320	-	Fiberboard: Sheet, Stock and Cut Shapes.
PPP-P-291	-	Paperboard, Wrapping, Cushioning.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.
PPP-T-495	-	Tubes, Mailing, and Filing.
QQ-S-781	-	Steel, Strapping, Flat.

MILITARY

MIL-C-215	-	Chemical Agent, White Phosphorus (WP).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:36Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

LD289028 - Projectile Load, Smoke, WP, MK14 Mod 0, List of Drawings, Sub-Assemblies and Parts.

US ARMY MUNITIONS COMMAND

D150-2-1 - Marking Diagram for Shipping Containers.

(All drawings referenced in this specification are included in or listed on one of the above drawings.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing LD289028 or on the subsidiary drawings, used in the filling and assembly of the projectile load, shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the projectile load shall conform to the specifications and drawings listed on Drawing LD289028 and subsidiary drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:36Z
Check the source to verify that this is the current version before use.

3.2 Filling and assembly.

3.2.1 Filling. The projectile load, clean and free of volatile corrosion inhibitor paper and foreign matter, shall be filled with molten WP, at a temperature of 130 ± 10°F, conforming to MIL-C-215 (see 6.2 and 6.3). When water is in contact with the WP during the filling operation, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. After test for leakage, the WP shall be solidified with the projectile in a filling end up position.

3.2.2 Filling height. The filling height shall be determined with the WP at a temperature of 130 ± 10°F with the filled and closed projectile in a vertical filling position. When properly filled, there shall be a void of 3 to 5 percent by volume remaining above the WP and water, when inspected in accordance with 4.2.4.1. The depth of water shall not exceed 1/8 inch. (See 6.3.)

3.2.3 Assembly. The projectile load shall be assembled and closed as shown on Drawing 1381153. Burster tube must be bottomed. (See 6.2.)

3.3 Cleaning. The exterior of the closed projectile load and the interior of the burster well shall be thoroughly cleaned to remove all grease, oil, and WP contamination (see 6.4).

3.4 Leakage. The cleaned and closed projectile load shall not leak when maintained at 160° ± 5°F for 8 hours and tested as specified in 4.2.4.2.

3.5 Workmanship. The projectile load shall be free from foreign matter (oil, grease, etc.) and damage such as dents, cracks, and deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:36Z
Check the source to verify that this is the current version before use.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Regular production.

4.2.1 Lotting. A lot shall consist of the projectile loads offered for acceptance at one time, which have been produced by one manufacturer, of one interfix number, at one plant, under essentially the same manufacturing conditions. However, no more than one manufacturer or interfix number of casings and burster wells shall be represented in any one lot of filled projectile loads.

4.2.2 Sampling. Sampling for examination and nondestructive tests shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection.

4.2.3.1 For examination and nondestructive test. Sample projectile loads shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each projectile load shall be inspected for the critical characteristics listed in the classification of defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:36Z
Check the source to verify that this is the current version before use.

4.2.3.3 Classification of defects.

(a) Projectile load, 5 inch 54 caliber, WP, MK14; WP filling assembly and marking (Dwg 1381153).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.2.4.2
2	WP contamination (3.3)	
Major:	AQL 1.0 percent defective	
*101	Marking incorrect	
*102	Filling excessive	4.2.4.1
*103	Water excessive	4.2.4.1
104	Burster well not properly bottomed	4.2.4.1
**105	Damage or internal contamination (3.5)	
Minor:	AQL 2.5 percent defective	
*201	Filling insufficient	4.2.4.1
202	Thread damaged	
203	Marking illegible	
204	Protective finish missing or incorrect	
205	Closure cap missing	
206	Foreign matter (3.5)	

* Note. Sampling for inspection for these characteristics after the first acceptable lot, shall be in accordance with MIL-STD-105, Level I.

** Internal contamination to be checked prior to filling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:36Z
Check the source to verify that this is the current version before use.

4.2.3.3 Classification of defects. (Continued.)

(b) Preparation for delivery (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 4.0 percent defective	
	101	Container incorrect
	102	Filler pads missing or insufficient to make a tight pack
	103	Container closure incorrect
	104	Strapping loose or missing
	105	Marking illegible or incorrect

4.2.4 Tests.

4.2.4.1 Radiographing. The filled and closed projectile load shall be radiographed while the WP is still molten as specified with the projectile in a vertical filling position and inspected to assure proper filling height and bottoming of the burster well in accordance with Drawing 1381153.

4.2.4.2 Leakage. The filled and closed projectile shall be placed in an oven in a filling-port-up position and maintained at the specified temperature for the required period of time and observed for evidence of leakage. The emission of smoke constitutes leakage.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. Six each empty projectile loads shall be packed in a class 2, style optional, nailed wood box, for type 3 load conforming to PPP-B-621. The internal box dimensions shall be 13 inches long by 8 3/4 inches wide by 16 5/8 inches high. The individual items shall be placed in a fiber tube 3 15/16 inches in diameter by 15 5/8 inches long, conforming to type I, class 1 or 2, style A, of PPP-T-495. As an alternate, the individual items may be placed in a full length open end, close fitting tube formed from paperboard conforming to type III, style 1 of PPP-P-291, with firmly taped seam using waterproof pressure sensitive tape conforming to class 3 of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:36Z
Check the source to verify that this is the current version before use.

PPP-T-60. Filler pads, fabricated from type CF or SF, weather resistant class fiberboard conforming to PPP-F-320, shall be placed between the sides, ends, top and bottom of the projectile load. Additional filler pads shall be used as required to make a tight pack. The box shall be closed in conformance with applicable requirements of PPP-B-621, using two each flat steel strapping 5/8 inch by 0.010 type I, class C per QQ-S-781.

5.2 Marking. In addition to any special marking required by the contract or order, shipping container shall be marked in conformance with Drawing D150-2-1.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Caution. Because of the hazardous nature of WP operations, filling installations are advised to take all necessary precautions required by proper industrial hygiene and safety practices to protect personnel from injury while filling projectiles in accordance with this specification.

6.3 Filling aid. Projectiles filled with approximately 6 3/4 pounds of WP have been found to comply with 3.2.2. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 Decontamination. Projectile loads may be decontaminated by passing them through a hot water and detergent spray washer or by flushing them with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well may be effected with a steam lance and a power driven wire brush.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 1320-A086

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:36Z
Check the source to verify that this is the current version before use.